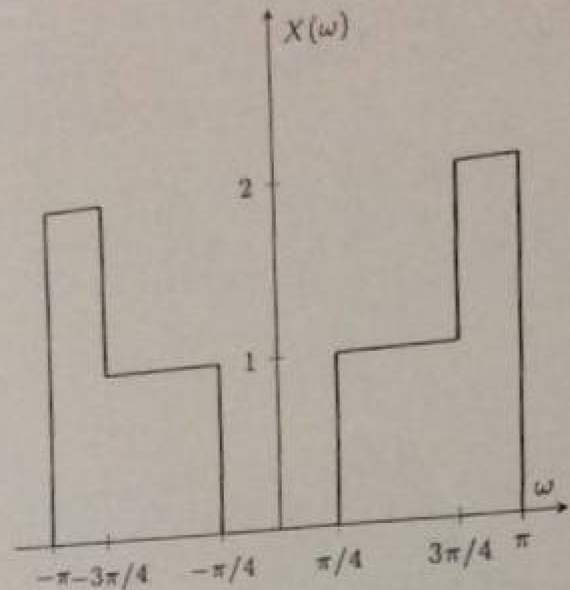

Figure 1: DTFT $X(\omega)$ for Problem 1 .

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Problems

1. (*Discrete-Time Fourier Transform*): Compute $x[n]$ for $X(\omega)$ as shown in Figure 1 . The final answer should be in terms of a real signal (without complex exponentials).

   Hint: You can use the following Fourier Transform relationship:

$$\frac{\sin\left(\frac{W}{2}n\right)}{\pi n} \xleftrightarrow{\text{DTFT}} \text{rect}\left(\frac{\omega}{W}\right)$$

2. (*Downsampling*): A discrete-time signal $x[n]$ has a Discrete-Time Fourier Transform (DTFT) $X(w)$ given by Figure 2.

$$2\pi\,\text{rect}\left(\frac{n}{W}\right) \longleftrightarrow \frac{\sin\left(-\frac{W}{2}w\right)}{-\frac{W}{2}w}$$

$-\frac{W}{2}tw$

$$2\pi\,\text{rect}\left(\frac{n}{W}\right)^{2/3} \longleftrightarrow$$

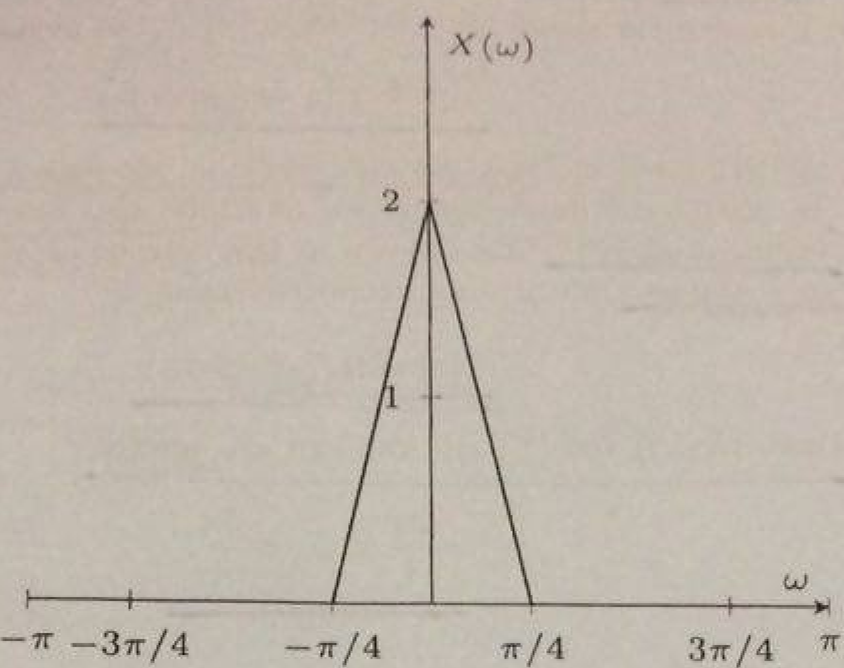

Figure 2: DTFT of $x[n]$ for Problem 2

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Let

$$y[n] = x[2n]$$

be the downsampled signal (downsampling by a factor of 2). State the expression for $Y(\omega)$ in terms of $X(\omega)$, and plot it as a function of ω (do not derive the expression for $Y(\omega)$).

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3. (*Single Sideband Modulation*):

(a) We know from the class lecture that the Complex Half-Band Filter has the impulse response given by

:(

$$h_{\mathbf{CHBF}}[n] = e^{j\frac{\pi}{2}n}\frac{\sin\left(\frac{\pi}{2}n\right)}{\pi n}.$$

State $H_{\mathbf{CHBF}}(\omega)$, and draw its magnitude plot.

(b) Again, from the class notes, we note that the Hilbert Transformer is given by

$$h_{\mathbf{HT}}[n] = 2\frac{\sin\left(\frac{\pi}{2}n\right)}{\pi n}\sin\left(\frac{\pi}{2}n\right).$$

3/3

State $H_{\mathrm{HT}}(\omega)$, and draw its magnitude plot.

(c) Consider the signal $x[n]$ from Problem 2. Let us define

$$\tilde{x}[n] \triangleq x[n] * h_{\mathrm{CHBF}}[n].$$

As discussed in class, we call $\tilde{x}[n]$ the complex analytic signal. In digital communications, we multiply $\tilde{x}[n]$ with the complex exponential $e^{j\omega_c n}$ (also known as the carrier) and then transmit its real part. Hence the transmitted signal is

$$y[n] = \mathrm{Re}\{e^{j\omega_c n}\tilde{x}[n]\}.$$

Draw $|\tilde{X}(\omega)|$ and $|Y(\omega)|$. You can assume that

$$\frac{2\pi}{4} < \omega_c < \frac{3\pi}{4}.$$

8 2 $x(\omega)^{+}$

NATIONAL UNIVERSITY
OF COMPUTER & EMERGING SCIENCES
LAHORE

FAST

Course: DSP — Answer Sheet No. 54156

Student's Name: SYED KAUSAIN MEHDI — Signature: Kausain.

Roll No.: 15L-4557 — Section: EE-2 — Date: 13-04-18

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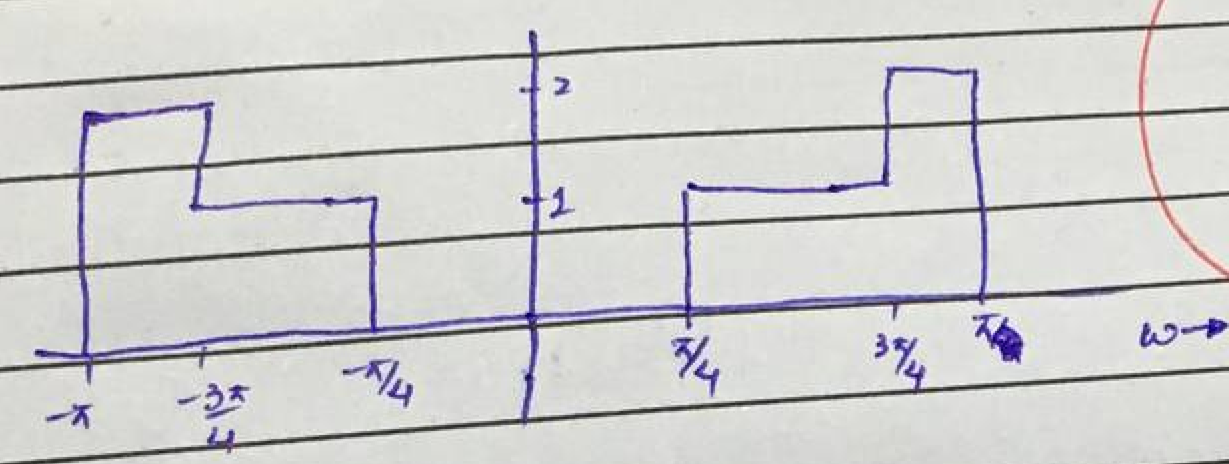
Q1



X(ω)

$x(n) = ?$

$$x(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\omega)\, e^{j\omega n}\, d\omega$$

30/30

: must be without exponentials:

$$X(\omega) = 2\,\text{rect}(\omega/2\pi) - \text{rect}\left(\omega/\tfrac{6\pi}{4}\right) - \text{rect}\left(\omega/\tfrac{2\pi}{4}\right)$$

$$\frac{\sin(W/2\, n)}{\pi n} \xrightarrow{\text{DTFT}} \text{rect}(\omega/W)$$

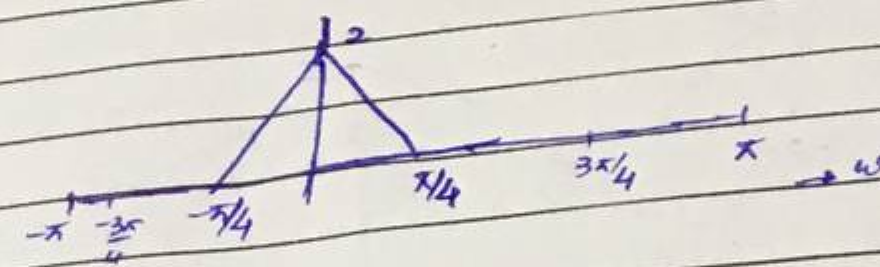
$$x(n) = 2\,\frac{\sin(\pi n)}{\pi n} - \frac{\sin(\frac{3\pi}{4} n)}{\pi n} - \frac{\sin(\frac{\pi}{4} n)}{\pi n}.$$

Rough Work

Part No.

Q_2 (Downsampling):

$X(\omega)$



$y[n] = x[2n]$

$$Y(\omega) = \sum_{n=-\infty}^{\infty} y[n]\, e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} x[2n]\, e^{-j\omega n}$$

Put $n' = 2n$; $n = n'/2$

$$= \sum_{n'=-\infty}^{\infty} x[n']\, e^{-\frac{j\omega n'}{2}}$$

$$\because \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = X(\omega)$$

$$Y(\omega) = X\left(\frac{\omega}{2}\right)$$

OR

$x[n] \longrightarrow \boxed{\downarrow D} \longrightarrow y[Dn]$

By using Downsampling equality/expression

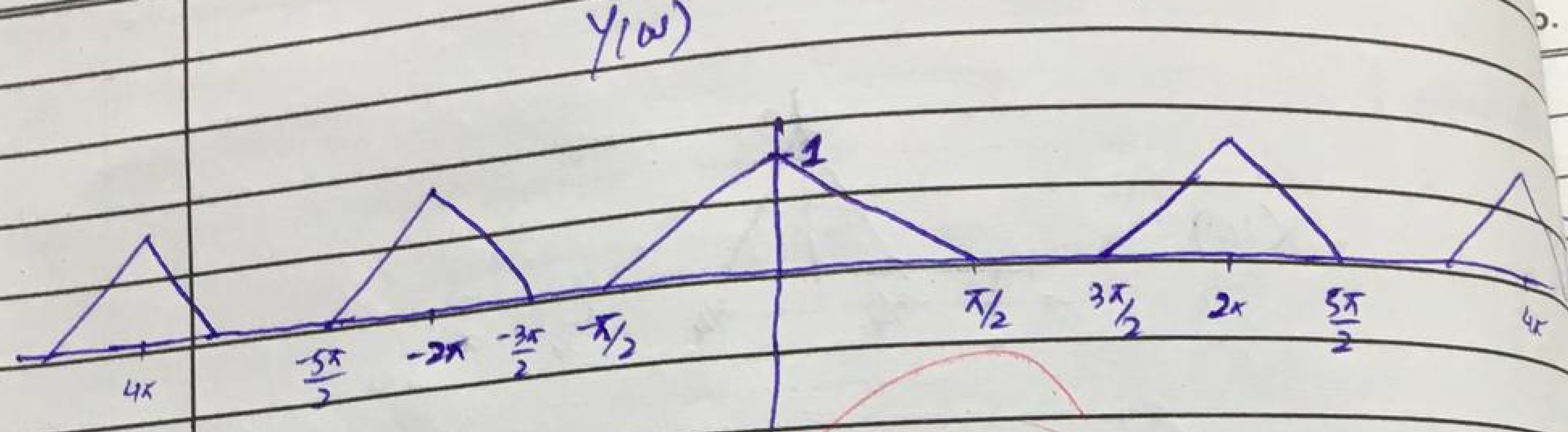
$$\frac{1}{D} \sum_{k=0}^{D-1} X\left(\frac{\omega - 2\pi k}{D}\right) \qquad \therefore D=2$$

$$Y(\omega) = \frac{1}{2} \sum_{k=0}^{1} X\left(\frac{\omega - 2\pi k}{2}\right)$$

$$= \frac{1}{2}\left[X\left(\frac{\omega}{2}\right) + X\left(\frac{\omega - 2\pi}{2}\right)\right]$$

$$= \frac{1}{2} X(\omega/2) + \frac{1}{2} X\left((\omega - 2\pi)/2\right)$$

Q / Part No.



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Complex Half-Band Filter

Q3

$$h_{CBHF}[n] = e^{j\frac{\pi}{2}n} \frac{\sin(\frac{\pi}{2} n)}{\pi n}$$

a)

$$\frac{\sin(\frac{\pi}{2} n)}{\pi n} \xleftrightarrow{DTFT} rect(\omega/\pi)$$

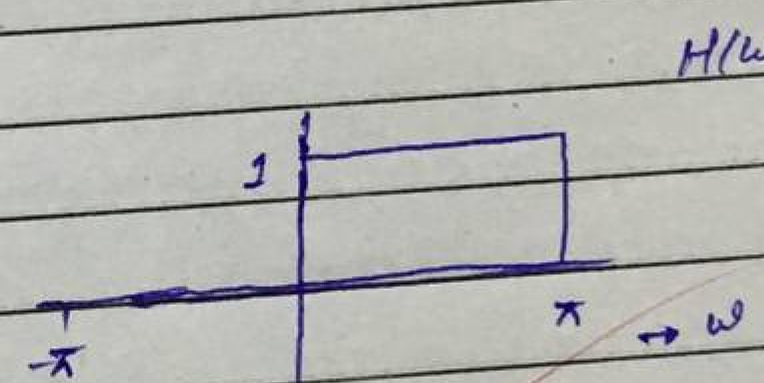
using property

$$\frac{\sin(W/2\, n)}{n\pi} \xleftrightarrow{DTFT} rect(\omega/W)$$

Hence

$$e^{j\frac{\pi}{2}n} \frac{\sin(\frac{\pi}{2} n)}{\pi n} \xleftrightarrow{DTFT} rect\left(\frac{\omega - \pi/2}{\pi}\right) \quad \because W = \pi$$

$$H_{CHBF}(\omega) = rect\left(\frac{\omega - \pi/2}{\pi}\right)$$



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b)

$$h_{HT}[n] = 2\frac{\sin(\frac{\pi}{2} n)}{\pi n} \sin(\frac{\pi}{2} n)$$

$$= 2\frac{\sin^2(\frac{\pi}{2} n)}{\pi n}$$

As $x(n) * \sin(\omega n) \xleftrightarrow{DTFT} \frac{e^{j\pi/2}}{2}\left[-X(\omega+\omega_c) + X(\omega-\omega_c)\right]$

Here

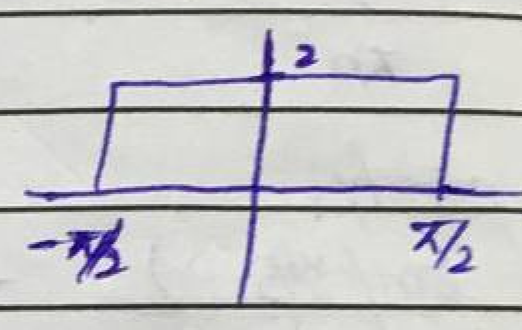
$$h_{HT}[n] = \left[2\frac{\sin(\frac{\pi}{2} n)}{\pi n}\right] \sin(\frac{\pi}{2} n)$$

Using Property

$$\frac{\sin(\frac{W}{2}n)}{\pi n} \xleftrightarrow{DTFT} rect\left(\frac{\omega}{W}\right)$$

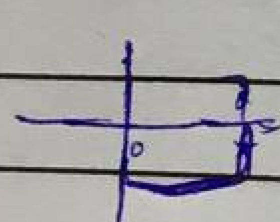
$$\frac{2\sin(\frac{\pi}{2}n)}{\pi n} \xleftrightarrow{DTFT} 2\, rect\left(\frac{\omega}{\pi}\right)$$

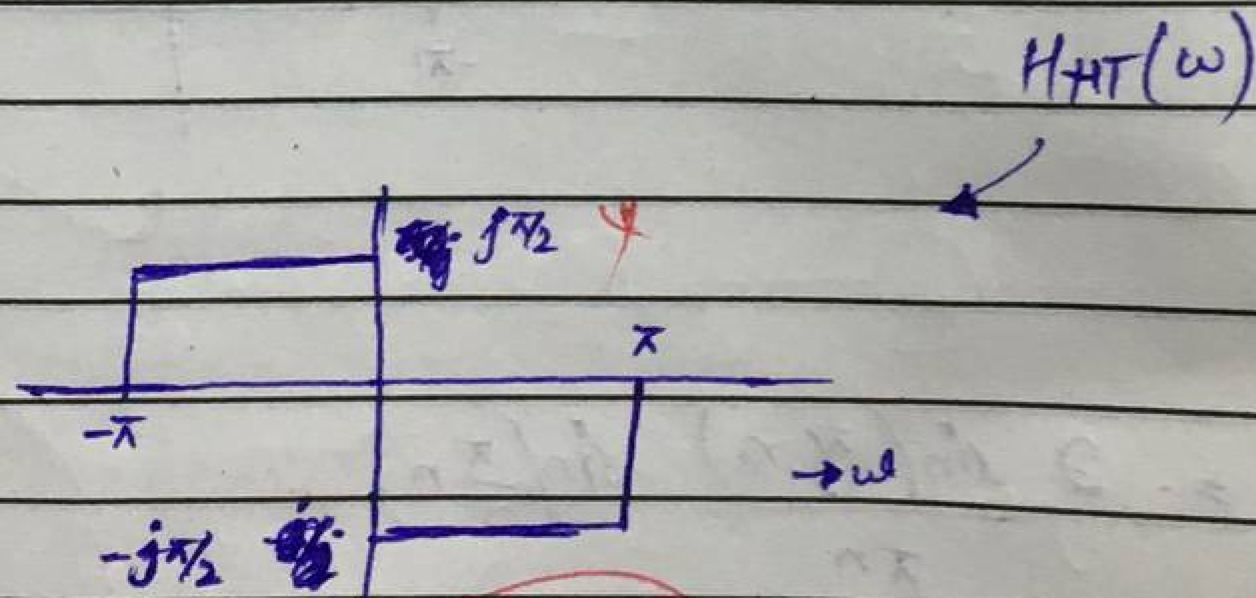
$$2rect\left(\frac{\omega}{\pi}\right)$$



$$\frac{2\sin(\frac{\pi}{2}n)}{\pi n}\cdot\sin\left(\frac{\pi}{2}n\right) \longleftrightarrow \frac{j\pi}{2j}\left[-rect\left(\frac{\omega-\pi/2}{\pi}\right)+rect\left(\frac{\omega+\pi/2}{\pi}\right)\right]$$

$$H_{HT}(\omega) = \frac{j\pi}{2}\left[-rect\left(\frac{\omega-\pi/2}{\pi}\right)+rect\left(\frac{\omega+\pi/2}{\pi}\right)\right]$$





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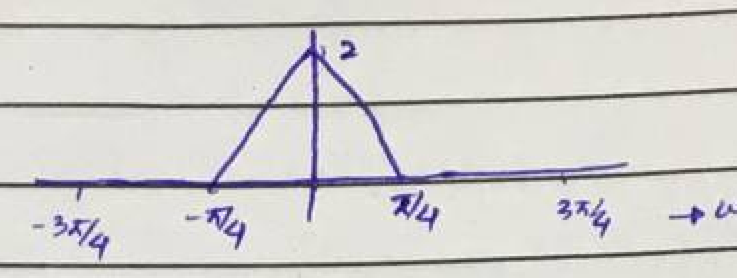
Q / Part No. — Rough Work

c)

$\tilde{x}[n] = x[n] * h_{CHBF}[n]$ → complex analytic signal

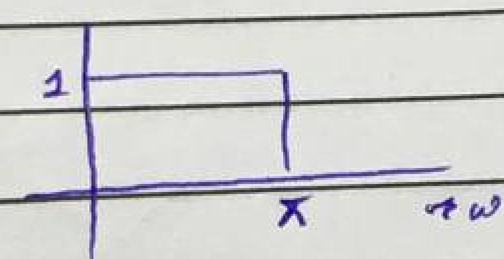
From problem-2 $X(\omega)$



$\tilde{X}(\omega) = X(\omega) \times H_{CHBF}(\omega)$ — (A)

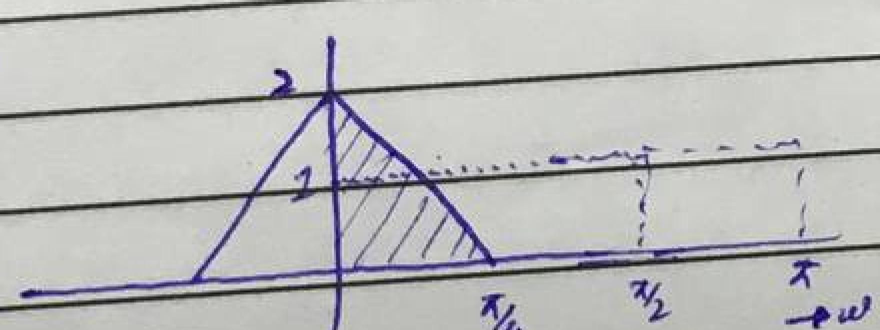
∵ Convolution in discretized domain ⟺ Multiplication in 'ω' domain

$H_{CHBF}(\omega) =$



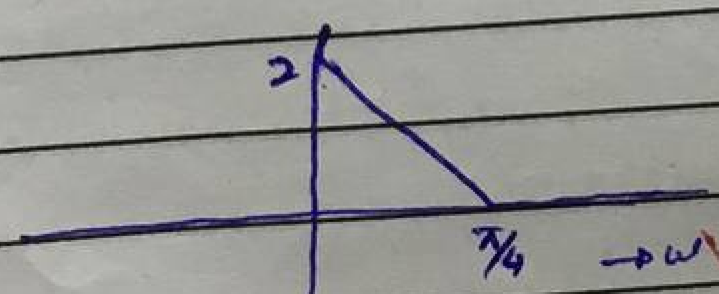
Hence from (A)

$\tilde{X}(\omega) = ?$



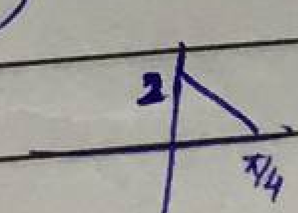
$|\tilde{X}(\omega)|$

↳



$y[n] = Re\{e^{j\omega_c n}\tilde{x}[n]\}$

$Re[\tilde{x}[n]]$



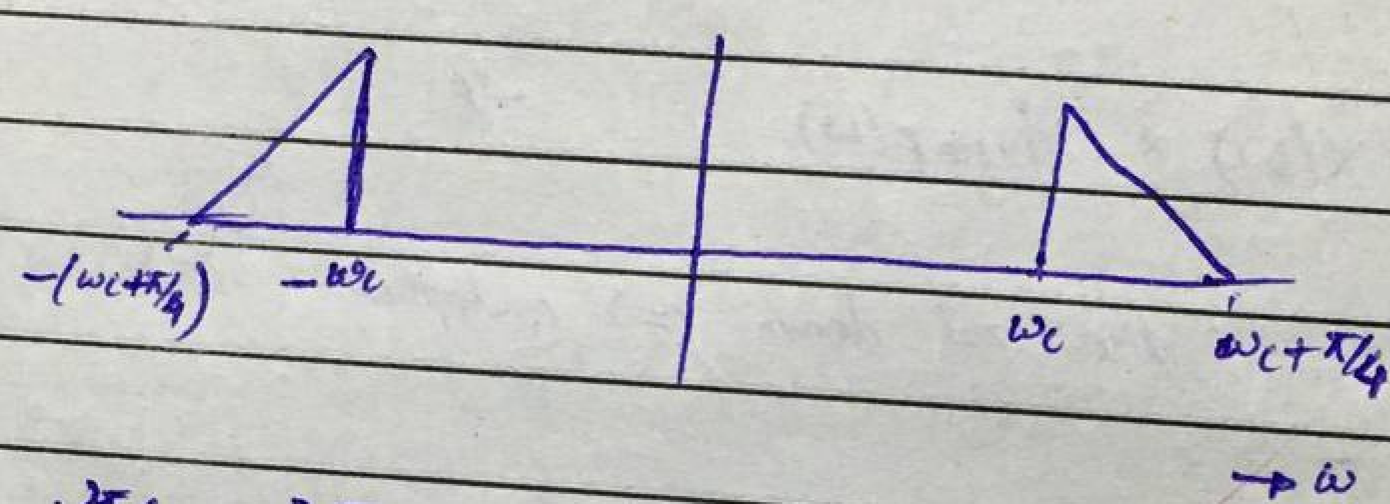
$[e^{j\omega_c n}] = \underbrace{\cos(\omega_c n)}_{Real} + j\underbrace{\sin(\omega_c n)}_{complex}$

$Re[e^{j\omega_c n}] = \cos(\omega_c n).$

Q / Part No.

$$Y(\omega) = \frac{1}{2}\left[\underset{\text{Real}}{X(\omega-\omega_c)} + \underset{\text{Real}}{X(\omega+\omega_c)}\right]$$

$|Y(\omega)|$



$\therefore \frac{2\pi}{4} < \omega_c < \frac{3\pi}{4}$